STILLSTACK · TRACEABILITY MATRIX

StillStack traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stillstack		
Source	docs/03b-traceability.md @ 4e01efb		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

StillStack traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 13 user requirements, 13 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-002	PR-001	Daily distillate	SSK-CAL-001, SSK-BLD-001, SSK-STM-001	Analysis (SSK-CAL-001 section 4), TRL 3; outdoor test beside a basin still at TRL 4	At risk: 10.3 L after warm-up (10.9 quasi-steady), 3 % margin, with the painted underside of D-07; 10.1 L quasi-steady at a glazing transmittance of 0.75; 13 L goal not met
UR-002	PR-002	Heat reuse	SSK-CAL-001, SSK-BLD-001, SSK-STM-001, SSK-OPT-001	Analysis (SSK-CAL-001 section 4), TRL 3	Met on paper: 1.29 quasi-steady, 1.22 after warm-up (1.7 % margin); 1.20 at a glazing transmittance of 0.75
UR-003	PR-003	Distillate salinity	SSK-CAL-001, SSK-BLD-001, SSK-STM-001	Analysis of carryover paths (SSK-CAL-001 section 7), TRL 3; conductivity meter per stage at TRL 4 (pass condition before any water is drunk)	To verify at TRL 4: brine carryover must stay below 8.2 mL/day (0.075 %)
UR-004	PR-004	Brine separation	SSK-CAL-001, SSK-BLD-001	Inspection of the model (`model.py --check`), TRL 3; conductivity per stage at TRL 4	At risk: 10 mm breaks at ribs, rails and plate edges, brine tails 10.5 mm over the closed lid and 44 mm from any distillate tongue, lip tabs on the high edges; carryover proven only by test
UR-005	PR-005	Passive operation	SSK-CAL-001, SSK-BLD-001, SSK-STM-001	Inspection of the design, TRL 3	Met: passive; trough rim at most 1.00 m (at 35 deg)
UR-006	PR-006	Feed water	SSK-CAL-001	Analysis (SSK-CAL-001 sections 6 and 7), TRL 3; wicking-rise test of the shortlisted fabrics	At risk: brine 66.7 g/L, far below saturation; wick permeability of 2.9e-11 m2 needed (3.0e-11 used for the test), not yet measured
UR-007	PR-007	Food-contact wetted parts	SSK-CAL-001, SSK-BLD-001	Review of the material list and certificates, TRL 3	At risk: fluoropolymer coating class chosen; no named product with a drinking-water certificate seen (rule of 2026-10-02: else 316 stainless for plates 1 and 2)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-008	PR-008	Dry stagnation	SSK-CAL-001, SSK-BLD-001, SSK-STM-001	Analysis (SSK-CAL-001 section 5) and datasheets, TRL 3	At risk: stages 1 and 2 reach 142 °C calm (8 K margin to the 150 °C rating, none at 1,100 W/m2; PPS rails long-term 218 °C); stage 3 reaches 107 °C calm, with PPS rails (SSK-DEC-001 D-01); inner glazing 113 °C against 120 °C (kept with the cover and shade rule as the barrier, D-05); wick and coating ratings not yet confirmed
UR-009	PR-009	Household scale	SSK-CAL-001, SSK-BLD-001, SSK-OPT-001	Analysis (SSK-CAL-001 section 8) and model, TRL 3; weigh at TRL 4	Met on paper: 18.8 kg dry (22.8 kg wet); footprint 1.28 x 1.25 m at 10 deg
UR-010	PR-010	Wick maintenance	SSK-CAL-001, SSK-BLD-001, SSK-STM-001, SSK-OPT-001	Design review (build plan SSK-BLD-001), TRL 3; timed at TRL 4	Met on paper: plates lift out once the low wall cap is off; wicks held by two 5052 aluminium clips each (D-08), no adhesive
UR-011	PR-011	Tilt range	SSK-CAL-001, SSK-BLD-001	Model check at every hole (`model.py --check`), TRL 3	Met: six prop holes, 10 to 35 deg in 5 deg steps
UR-012	PR-012	Cost and buildability	SSK-CAL-001, SSK-OPT-001	Priced BOM (SSK-CAL-001 section 10), TRL 3	Over target: USD 460.70 (USD 105.70 over), after the stage 3 PPS rails of SSK-DEC-001 D-01 (USD 16)
UR-013	PR-013	Wind and hot surfaces	SSK-CAL-001, SSK-BLD-001, SSK-STM-001	Analysis (SSK-CAL-001 section 9), TRL 3; stake pull-out in soft or sandy ground at TRL 5	Met on paper anchored: with 40 kg of ballast it tips at 28.8 m/s, and each rear stake needs about 6 N of hold-down at 17 m/s; unanchored it tips at 16.5 m/s (lift at 18.7 m/s), which the hold point excludes; cover BOM line 14, panel label BOM line 19

Gaps

None.